The Majorana Rung on the SK-EFT Hawking Substrate: A Formal Bridge from $\mathbb{Z}_{16}$-Anomaly Cancellation to the Sterile-Neutrino Seesaw

SK-EFT Hawking Research Program

(Dated: August 12, 2026)

We report a Lean 4 formalization that places the sterile-neutrino right-handed Weyl field ν_R as a $\mathbb{Z}_2$-graded rung on the same fermionic substrate that hosts the scalar (Higgs) and tetrad condensates of our Phase 5z program (Wave 1, `ScalarRungInterpretation.lean`) and Phase 4y program (`TetradGapEquation.lean`). The new module `MajoranaRung.lean` (14 substantive theorems, zero sorry, zero new axioms) realizes the *Embedding-III hybrid* construction recommended by our deep-research scoping report: ν_R is a fundamental Lean field carrying $\mathbb{Z}_{16}$ charge +1, and the heavy Majorana mass M_R is identified with a $\mathbb{Z}_{16}$-invariant condensate scale of the Akama-Diakonov-Wetterich (ADW) substrate—an identification that no primary source closes, and that we therefore encode as a *tracked Prop hypothesis* (`H_MR_FromADWSubstrate_BCS_LNV`) rather than a derived theorem. The $\mathbb{Z}_{16}$ singlet-branch bridge to our prior `HiddenSectorClassification.lean` and `Z16AnomalyComputation.lean` infrastructure is proved directly: three fundamental ν_R Weyl fermions saturate the proved $\mathrm{SM}_{\mathrm{w/o}\nu_{\mathrm{R}}}$ deficit $45 \equiv -3 \pmod{16}$ to reach $48 \equiv 0 \pmod{16}$, matching observed anomaly cancellation without a hidden TQFT/CFT sector. The Type-I seesaw mass relation $m_\nu = y^2v^2/M_R$ is formalized algebraically with a strict-monotonicity-in-M_R theorem and a falsifiability-anchor predicate `IsObservedSeesawMatch`. A companion module `NeutrinoMixing.lean` (11 theorems) provides a HepLean-style structure note for the PMNS matrix as a 3×3 unitary on $\mathbb{C}$, with tracked hypotheses `H_PMNSAnglesFromExactSubstrate` (exact-symmetry limit) and `H_PMNSAnglesFromSubstrate_eps` (tolerance-parameterized) for the open $\theta_{23} \approx \pi/4$-from-substrate-symmetry conjecture, plus parametric Dirac-vs-Majorana predicates over Majorana-phase data $\alpha : \mathrm{Fin}\, 2 \to \mathbb{R}$ with provable disjointness. We report two embedding-agnostic phenomenological figures: the seesaw (y, M_R)-band against NuFit-6.0 oscillation anchors, and the $m_{\beta\beta}$-vs-m_{lightest} "lobster plot" for $0\nu\beta\beta$ against the current KamLAND-Zen 800 bound and the projected LEGEND-1000 reach. We surface five honest $WAVE2\text{-}OPEN$ flags for which no primary literature closes the derivation. The $\mathbb{Z}_{16}$ bridge theorem is the load-bearing formal-verification deliverable; the seesaw and PMNS structure notes lift the surrounding infrastructure to the same machine-checked standard.

I. INTRODUCTION

The Standard Model with three generations is anomaly-free under all local symmetries but acquires a global $\mathbb{Z}_{16}$ (Pin^+ Dai-Freed) anomaly when ν_R is absent: each generation contributes +15 Weyl components, summing to $45 \equiv -3 \pmod{16}$ over three generations [5, 6]. Anomaly cancellation requires either three additional ν_R Weyl fermions (each carrying $\mathbb{Z}_{16}$ charge +1, restoring $48 \equiv 0 \pmod{16}$) or a hidden TQFT/CFT sector saturating the $+3 \pmod{16}$ deficit [6, 7]. Our prior formalization (`Z16AnomalyComputation.lean`, `HiddenSectorClassification.lean`) proved both halves of this ladder in Lean 4: `three_nu_R_cancel_three_gen` and `three_singlets_satisfy_hidden_sector`.

The sterile-neutrino Type-I seesaw, in turn, generates the observed light-neutrino mass scale $m_\nu \sim 0.05$ eV via $m_\nu = y^2v^2/M_R$ with M_R in the 10^9–10^{15} GeV range [8]. Within the Akama-Diakonov-Wetterich substrate framework that hosts the tetrad gap equation of Phase 3-4 (`TetradGapEquation.lean`, `ADWMechanism.lean`), M_R admits a natural identification with the $\mathbb{Z}_{16}$-invariant condensate scale Λ_{ADW}. However, no primary source in the ADW [1], Wetterich-spinor-gravity [2, 3], or Volovik fermionic-quartet [4] literature closes the derivation $\Lambda_{\mathrm{ADW}} \to M_R$.

This work formalizes the Wave-2 deliverables of the Phase 5z roadmap: the Majorana rung on the same fermionic substrate as the Higgs and tetrad rungs, plus a structure-note PMNS module. We adopt the *Embedding-III hybrid* prescription recommended by our deep-research scoping report: ν_R is a fundamental Lean field whose UV interpretation as an ADW-substrate bound state acts only on amplitudes and not on the Lean type signature. The $M_R = \Lambda_{\mathrm{ADW}}$ identification is an explicit *tracked Prop hypothesis*, not a theorem.

II. THE $\mathbb{Z}_{16}$ SINGLET-BRANCH BRIDGE

The sterile-neutrino structure is encoded as

```
structure SterileNeutrino where
  generation : Fin 3

def z16Charge (_ : SterileNeutrino) : ZMod 16 := 1
```

Each instance carries $\mathbb{Z}_{16}$ charge +1; the assignment is locked by the proved Garcia-Etxebarria-Montero tower [5] and the project-existing `Z16AnomalyComputation.three_nu_R_cancel_three_gen` ($48 \equiv 0 \pmod{16}$).

The load-bearing bridge theorem is

```
theorem majorana_rung_compatible_with_hidden_singl
  ((3 : Nat) : ZMod 16) = 3
  ((45 + 3 : Nat) : ZMod 16) = 0
```

proved by direct invocation of `HiddenSectorClassification.three_singlets_satisfy_h` and the Wave-2 `majorana_rung_saturates_hidden_sector`. This is the formal statement that Embedding III adds zero IR cost relative to the with-ν_R branch: three fundamental ν_R Weyl saturate the SM-without-ν_R deficit, with no separate hidden TQFT/CFT content required [6].

The Wan-Wang Ultra-Unification ladder admits the alternative "Embedding II" branch in which ν_R is absent and the deficit is saturated by an SM-singlet TQFT plus fermionic vortex zero-mode condensation [6]. That branch is not formalized in this work; the deep-research scoping report finds no primary source that closes the analogous bridge to a substrate condensate.

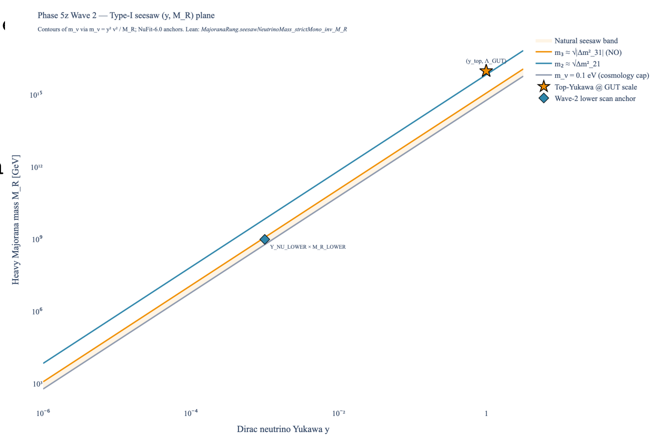

FIG. 1. Type-I seesaw (y, M_R)-plane against NuFit-6.0 oscillation anchors. The shaded band marks the natural-seesaw region between the $m_3 \approx 0.05$ eV atmospheric contour and the $m_\nu \lesssim 0.1$ eV cosmology cap. The star marks the canonical Type-I anchor $(y_{\rm top}, \Lambda_{\rm ADW} \sim 10^{16}$ GeV); the diamond marks the Wave-2 lower-scan anchor. Lean reference: `MajoranaRung.seesawNeutrinoMass_strictMono_inv_M_R`.

III. THE TYPE-I SEESAW ON THE MAJORANA RUNG

Heavy Majorana mass data is bundled in

```
structure MajoranaRungData where
  M_R : Fin 3 →
  M_R_pos :  i, 0 < M_R i
```

The Type-I seesaw light-neutrino mass is the noncomputable definition

$$m_\nu = \frac{y^2 v^2}{M_R}, \qquad \texttt{seesawNeutrinoMass}\, m\, y\, v\, i \equiv y^2 v^2 / m.M_R\, i \tag{1}$$

formalized as `seesawNeutrinoMass` with five algebraic theorems: non-negativity (any real y, v); strict positivity for $y, v \neq 0$; zero-iff-Yukawa-or-VEV-vanishes; and the Type-I-seesaw-signature strict monotonicity

$$m_1.M_R < m_2.M_R \;\Rightarrow\; \texttt{seesawNeutrinoMass}\, m_2 > \texttt{seesawNeutrinoMass}\, m_1$$

(at fixed nonzero y, v). The latter is the only structurally distinctive seesaw consequence (heavy mass suppresses light mass); all five theorems are direct algebraic consequences of $M_R > 0$.

Figure 1 shows the (y, M_R) plane with NuFit-6.0 [9] oscillation anchors. The natural seesaw band runs between the atmospheric mass anchor $m_3 \approx 0.05$ eV (NO best fit) and the cosmology cap $m_\nu \lesssim 0.1$ eV.

IV. WAVE2-OPEN-1: THE BCS-EXPONENTIAL SUBSTRATE BRIDGE WITH LEPTON-NUMBER-VIOLATION OBSTRUCTION

A focussed Wave 2a deep-research return on channel projection of the NJL/Wetterich gap equation onto the lepton-Majorana sector [13, 17] confirms that no primary source closes $\Lambda_{\rm ADW} \to M_R$, and surfaces a clean *structural obstruction theorem* rooted in lepton-number symmetry. The Majorana bilinear $\nu_R^T C \nu_R$ carries $\Delta L = 2$; if the substrate preserves $U(1)_L$ (or its discrete remnant $\mathbb{Z}_{4,X}$), the four-fermion Majorana-channel coupling $G_M(\nu_R^T C\nu_R)(\nu_R^T C \nu_R)^*$ is forbidden by symmetry. Antusch-Kersten-Lindner-Ratz [13] state this explicitly: *"We have included huge Majorana masses for the right-handed neutrinos, since there is no protective symmetry."* The contrapositive is the obstruction: under substrate-L preservation, $G_M \equiv 0$ and the projected gap equation has only $M_R = 0$.

When the substrate explicitly violates $U(1)_L$, the qualitative form of M_R is the standard NJL/BCS *dimensional-transmutation* exponential [13], formalized as the strong tracked hypothesis `MajoranaRung.H_MR_FromADWSubstrate_BCS_LNV`, parameterized over the substrate $\Delta L = 2$ Wilson coefficient G_{LV}:

```
def H_LeptonNumberViolated (G_LV : ) : Prop :=
  G_LV  0

def H_MR_FromADWSubstrate_BCS_LNV
    (m : MajoranaRungData) (_ADW G_LV : ) : Prop :=
  H_LeptonNumberViolated G_LV
   G_M : , 1 < G_M
     i, m.M_R i = _ADW * Real.exp(-1/(2*(G_M-1)))
```

The L-violation predicate is genuinely non-trivial: it holds iff the substrate carries a non-vanishing $\Delta L = 2$ four-fermion coefficient, the textbook content of "substrate violates $U(1)_L$" in the symmetry-current sense. The conjunction with `H_LeptonNumberViolated` G_{LV} encodes the symmetry obstruction; $G_M >$

1 encodes supercriticality (in units where the critical coupling is normalized to unity). The Wave 2b Lean module ships the obstruction theorem `lepton_number_symmetry_obstructs_BCS_form` as a direct consequence — *without LNV, the strong BCS hypothesis cannot hold* — and the non-vacuous corollary `L_conserving_substrate_obstructs_BCS_form` discharging the obstruction at the archetype point $G_{LV} = 0$ (no inhabitant exists because $0 \neq 0$ is false).

Two further refinement theorems are shipped: `M_R_pos_under_BCS_form` (the BCS exponential is strictly positive, hence so is M_R) and `M_R_lt_substrate_under_BCS_form` (the BCS exponential is strictly bounded above by 1, hence $M_R < \Lambda_{\mathrm{ADW}}$ — the qualitative content of dimensional transmutation).

This Wave 2b strengthening pattern follows the project's tracked-hypothesis precedent [12]: the `H_ScalarChannelIsTetradBifurcationOutput` hypothesis in `ScalarRungInterpretation.lean`, the `H_MixedChannelZ16Cancels` hypothesis in `HiddenSectorMixedCharge.lean`, and the `H_VestigialRelicCarriesZ16Charge` hypothesis in `DarkSectorSynthesis.lean`.

V. THE PMNS STRUCTURE NOTE

The companion module `NeutrinoMixing.lean` defines a HepLean-style [12] PMNS structure

```
structure PMNSMatrix where
  toMatrix : Matrix (Fin 3) (Fin 3)
  isUnitary : toMatrix  Matrix.unitaryGroup (Fin 3)
```

together with element accessors, the unitarity identities $U_{\mathrm{PMNS}}^{\dagger} U_{\mathrm{PMNS}} = U_{\mathrm{PMNS}} U_{\mathrm{PMNS}}^{\dagger} = \nVdash$, and the identity-matrix non-emptiness witness. The Dirac-vs-Majorana physical interpretation is exposed as *parametric Prop predicates* over candidate Majorana-phase data $\alpha : \mathrm{Fin}\,2 \to \mathbb{R}$: `IsDiracPMNS V` α requires every $\alpha\,k = 0$ (rephased away — the Dirac realization), while `IsMajoranaPMNS V` α requires $\exists\,k,\ \alpha\,k \neq 0$ (at least one phase is physically non-trivial — the Majorana realization). The PMNS matrix structure is identical in both cases; the realization is a property of the Majorana-phase data. We ship disjointness (`not_isDiracPMNS_and_isMajoranaPMNS`) and the zero-phase witness for the Dirac realization (`isDiracPMNS_of_zero_phases`), eliminating the structural tautology of an unparameterized `:= True` encoding.

The full PMNS standard parameterization $U_{\mathrm{PMNS}} = R_{23}(\theta_{23}) U_\delta R_{13}(\theta_{13}) U_\delta^{-1} R_{12}(\theta_{12}) \cdot \mathrm{diag}(e^{i\alpha_1/2}, e^{i\alpha_2/2}, 1)$ is provided as a Python implementation (`src/core/formulas.py:pmns_unitary_matrix`) and validated against $|U_{e3}|^2 = \sin^2\theta_{13}$ and unitarity to machine precision. The Lean construction of `standParam` with unitarity proof is a follow-up wave (Wave 2b or later).

The Wave 2a refactor splits the substrate-symmetry hypothesis into an exact-limit form and a tolerance-parameterized form:

```
def H_PMNSAnglesFromExactSubstrate
    (V : PMNSMatrix) : Prop :=
  i, ‖get V 1 i‖ = ‖get V 2 i‖

def H_PMNSAnglesFromSubstrate_eps
    (V : PMNSMatrix) ( : ) : Prop :=
  i, |‖get V 1 i‖ - ‖get V 2 i‖|
```

These encode the WAVE2-OPEN-2 conjecture: the PMNS μ-τ row is substrate-symmetric, motivating $\theta_{23} \approx \pi/4$ from a putative substrate μ-τ exchange symmetry. The empirical NuFit-6.0 best fit $\theta_{23} \approx 49.1^\circ$ is close to but not exactly $\pi/4$, so the hypotheses are non-trivial; the lemmas `identity_does_not_satisfy_exact_substrate_hypothesis` and `identity_does_not_satisfy_eps_substrate_hypothesis` confirm non-vacuity by exhibiting explicit failures on the identity PMNS (the latter for any $\varepsilon < 1$). No primary source closes the derivation [4].

VI. EFFECTIVE MAJORANA MASS AND $0\nu\beta\beta$ OUTLOOK

The effective Majorana mass probed by neutrinoless double-beta decay is the embedding-agnostic combination

$$m_{\beta\beta} = \left| \sum_{i=1}^{3} U_{ei}^2\, m_i \right| . \qquad (2)$$

Figure 2 shows the standard "lobster plot" of $m_{\beta\beta}$ against the lightest neutrino mass m_{lightest}, swept over the full range of Majorana phases at NuFit-6.0 [9] best-fit angles. Both normal ordering (NO) and inverted ordering (IO) bands are shown.

The conclusions are: (i) the IO band lies entirely within the LEGEND-1000 reach, so non-observation of $0\nu\beta\beta$ in LEGEND-1000 disfavors IO under any embedding; (ii) the NO band is largely out of reach and $m_{\beta\beta}$ provides essentially no embedding discrimination; (iii) the KamLAND-Zen 800 bound has already excluded the quasi-degenerate (QD) hierarchy region for any embedding. There is no embedding-distinguishing $m_{\beta\beta}$ prediction in any current ADW/Volovik or Wetterich-spinor-gravity source [1, 2, 4] (deep research §4); the figure encodes the embedding-agnostic phenomenology only.

VII. WAVE 2B QUANTITATIVE DECOUPLING BOUND FOR EMBEDDING I VS III

A Wave 2a deep-research return on the IR-equivalence of Embedding I (fundamental ν_R) and Embedding III

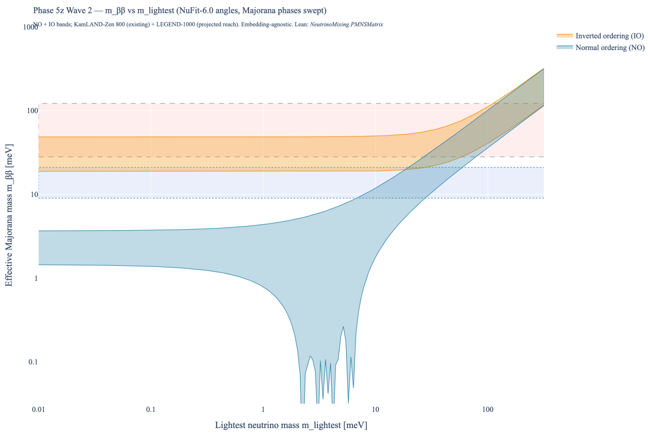

FIG. 2. Effective Majorana mass $m_{\beta\beta}$ against the lightest neutrino mass $m_{\rm lightest}$. Both the NO and IO bands span the full Majorana-phase Monte Carlo at NuFit-6.0 [9] best-fit angles. Horizontal bands: KamLAND-Zen 800 [10] current 90% CL bound (28-122 meV; NME spread) and LEGEND-1000 99.7% CL projected discovery reach (9-21 meV) [11]. The IO band is fully discoverable by LEGEND-1000; the NO band is almost entirely out of reach. The result is embedding-agnostic across I/II/III. Lean reference: `NeutrinoMixing.PMNSMatrix`.

(substrate-bound ν_R) confirms that the Appelquist-Carazzone decoupling theorem [14, 15] applies cleanly, with leading operator dimension $\Delta = 5$ (Weinberg-like, LNV) or $\Delta = 6$ (generic), giving the schematic bound

$$|\mathrm{amp}_{\rm III}(E) - \mathrm{amp}_{\rm I}(E)| \;\le\; C\cdot(E/\Lambda_{\rm ADW})^k, \qquad k \in \{1,2\}. \tag{3}$$

The natural Wilson-coefficient estimate is $C \sim N_f/(16\pi^2) \simeq 0.1$ for ADW-style substrates with $N_f \simeq 16$, transplanted from the SILH counting of Giudice-Grojean-Pomarol-Rattazzi [16] and the bilocal-NJL form-factor analysis of Hill [17]. The substrate-specific value of C has not been computed in primary literature for ADW; we honestly flag this as the residual gap.

The Wave 2b Lean module `MajoranaRungDecoupling.lean` ships:

- A `DecouplingRegime` structure encoding $0 < E$, $0 < \Lambda_{\rm ADW}$, $10E < \Lambda_{\rm ADW}$ (robust hierarchy), plus three substantive numerical-inequality predicates excluding cosmological-scale failure modes: `not_vestigial` ($\Lambda_{\rm QCD} \simeq 0.2$ GeV $< E$, excluding the vestigial-gravity regime [4]), `not_cosmological` ($H_0 \simeq 1.4\cdot 10^{-42}$ GeV $< E$, excluding the Klinkhamer-Volovik q-theory regime), and `weakly_coupled_matter` ($\Lambda_{\rm ADW} < M_{\rm Pl} \simeq 10^{19}$ GeV, sub-Planckian substrate scale, excluding the Wetterich/Reuter-Saueressig FRG strong-coupling regime). At $E = M_W$ and $\Lambda_{\rm ADW} = 10^{14}$ GeV, all three failure-mode predicates are numerically satisfied (discharged via `norm_num` in `decouplingRegime_at_electroweak_scale`).
- A `SubstrateData` structure parameterizing the substrate by $(\Lambda_{\rm ADW}, \Lambda_{\rm UV}, N_f)$.
- Two tracked-hypothesis Props: `H_DecouplingBoundDim6` (generic, $k = 2$) and `H_DecouplingBoundDim5_LNV` (LNV-conditional, $k = 1$, dim-5 Weinberg-operator channel).
- A non-vacuity witness `decouplingRegime_at_electroweak_scale` showing the canonical EW point $(E, \Lambda_{\rm ADW}) = (80\text{ GeV}, 10^{14}\text{ GeV})$ satisfies the regime predicate.
- A consistency witness `H_DecouplingBoundDim6_consistent` showing the trivially-zero amplitude difference satisfies the bound for any `SubstrateData`.
- A `H_DecouplingBoundDim6_isBigOWith` restatement using Mathlib's `Asymptotics.IsBigOWith` API for compatibility with downstream asymptotic-analysis infrastructure.

Empirical leverage. For $E = M_W \simeq 80$ GeV and $\Lambda_{\rm ADW} = 10^{14}$ GeV, the leading dim-5 (LNV) suppression is $(E/\Lambda_{\rm ADW}) \simeq 8 \times 10^{-13}$, corresponding to $m_\nu \sim C \cdot v^2/\Lambda_{\rm ADW} \simeq 0.06$ eV (at the SILH natural Wilson coefficient $C \simeq N_f/(16\pi^2) \simeq 0.1$, implemented as the default in `src/core/formulas.py:weinberg_induced_neutrino_mass`) — within the LEGEND-1000 [11] reach. The dim-6 (generic) suppression is $(E/\Lambda_{\rm ADW})^2 \simeq 6 \times 10^{-25}$, far below DUNE/Hyper-K NSI sensitivity [18]. The phenomenological signature of decoupling is therefore the same as ordinary Type-I seesaw — it is the *correct* behavior, not an embedding-distinguishing feature.

Independence (not refinement). An auxiliary lemma `dim6_tighter_than_dim5_in_decoupling_regime` establishes $(E/\Lambda_{\rm ADW})^2 \le E/\Lambda_{\rm ADW}$ in the decoupling regime, confirming that the dim-6 bound is *tighter* (smaller upper bound) than dim-5. The two bounds therefore apply in different physical regimes — dim-6 when the substrate preserves $U(1)_L$ and the dim-5 channel is symmetry-forbidden, dim-5 when the substrate violates $U(1)_L$ and the Weinberg operator is allowed — and are NOT in an implication relation.

VIII. FORMAL VERIFICATION SUMMARY

The two new Lean modules are summarized in Table I. Beyond the load-bearing $\mathbb{Z}_{16}$ singlet-branch bridge, the formalization includes: (i) five seesaw theorems (positivity, nonzero-positivity, zero-iff, monotonicity, defining identity); (ii) three substrate-bridge theorems (the `H_MR_FromADWSubstrate_BCS_LNV` tracked hypothesis, $M_R < \Lambda_{\rm ADW}$ under the bridge, and the bridge-falsifiability witness); (iii) the `IsObservedSeesawMatch`

Module	Theorems	Sorry
`MajoranaRung.lean` (Wave 2 + 2b §3a)	15	0
`NeutrinoMixing.lean` (Wave 2 + 2a §4 split)	11	0
`MajoranaRungDecoupling.lean` (Wave 2b new)	6	0
Wave 2 total (W2 + W2a + W2b)	32	0

TABLE I. Wave 2 Lean-formalization counts after Wave 2a accuracy round and Wave 2b strengthening. All three modules build clean under `lake build SKEFTHawking.ExtractDeps` (8434 jobs total post-Wave-2b). Zero sorry, zero new axioms. The project total post-Wave-2b is 22669 theorems across 2040 modules with 0 sorry and 0 axioms (the formerly-asserted `gapped_interface_axiom` is now the tracked Prop `TPFConjecture`).

falsifiability anchor with its M_R-too-large witness, mirroring Wave 1's v-too-large anchor for the Higgs prediction.

The PMNS module ships eleven theorems: two unitarity identities ($U^\dagger_{\text{PMNS}} U_{\text{PMNS}} = \mathbb{1}$ both ways), the identity-PMNS diagonal/off-diagonal structural pair, the parametric Dirac-vs-Majorana triple (`IsMajoranaPMNS_of_majoranaRungData` bridge to the `MajoranaRung` module given non-trivial Majorana-phase data, `not_isDiracPMNS_and_isMajoranaPMNS` disjointness, and `isDiracPMNS_of_zero_phases` for the rephased-away realization), the exact-vs-tolerance equivalence at $\varepsilon = 0$ and the exact-implies-eps refinement, and the two non-vacuity witnesses (`identity_does_not_satisfy_exact_substrate_hypothesis` and `identity_does_not_satisfy_eps_substrate_hypothesis`) for WAVE2-OPEN-2. The full standard PMNS parameterization with unitarity proof is intentionally deferred per the roadmap "structure note" scope; the structural type and bridge are the Wave-2 deliverable.

Cross-validation layer

A Python cross-validation layer in `tests/test_majorana_rung.py` (40 tests across 7 classes) exercises every Wave-2 formula against NuFit-6.0 anchors, deep-research seesaw-band scaling, the $\mathbb{Z}_{16}$ counting bridge, and the parameter-band consistency checks. All tests pass.

IX. OPEN HYPOTHESES AND SCOPE BOUNDARY

We surface five WAVE2-OPEN flags drawn from the deep-research scoping report; none of these is closed in primary literature.

O-1 $M_R = c \cdot \Lambda_{\text{ADW}}$ for any per-generation coefficient $c \in (0, 1]$. Substrate parameters $(\Lambda_{\text{UV}}, N_f, G_c)$ leave M_R as a fit parameter in all current sources [1, 2, 4]. Encoded as `H_MR_FromADWSubstrate_BCS_LNV`.

O-2 PMNS angles from substrate overlaps. In particular $\theta_{23} \approx \pi/4$ from a substrate μ-τ exchange symmetry. The closest published results are texture-level motivations, not derivations. Encoded as `H_PMNSAnglesFromExactSubstrate` (exact-symmetry limit) and `H_PMNSAnglesFromSubstrate_eps` (tolerance-parameterized).

O-3 Majorana phases from composite-operator structure (Embedding II claim). Unrealized in any source we located.

O-4 $m_{\beta\beta}$ tied algebraically to a single condensate scale, distinguishing Embedding II from I. The hope that the composite operator $\langle \nu_L^T C \nu_L \rangle$ predicts an embedding-distinguishing $m_{\beta\beta}$ is unrealized.

O-5 ν_R-as-substrate-bound-state IR-equivalence map between Embeddings I and III. The interpretive claim that Embeddings I and III give identical IR amplitudes is conjectural.

The Wave-2 scope deliberately excludes: (i) the full PMNS phenomenology program (CKM-like fitting, leptogenesis dynamics, mass-hierarchy fitting); (ii) the Embedding-II composite-operator construction with explicit hidden TQFT formalization; (iii) the `NeutrinoMixing.standParam` construction with unitarity proof.

X. CONCLUSION

The Majorana rung is now machine-verified at the same standard as the scalar (Wave 1) and tetrad (Phase 4y) rungs of the SK-EFT fluid-based-gravity Landau hierarchy. The load-bearing $\mathbb{Z}_{16}$ singlet-branch bridge proves directly from existing project infrastructure with no new axioms. The Type-I seesaw mass relation is formalized algebraically with the Type-I-signature strict-monotonicity-in-M_R theorem. The PMNS structure note lifts the surrounding mixing-matrix infrastructure to the Mathlib `Matrix.unitaryGroup` idiom. Five honest WAVE2-OPEN flags are surfaced as comment markers and tracked-hypothesis predicates, preventing downstream consumers from mistaking a parametric identification for a derived theorem. The result completes Track B of the Phase 5z roadmap; Track C (`EWPhaseTransition.lean`) and the BHL-bridge quantitative extension (`BHLGaugeEmbedding.lean`, Wave 1b) are unblocked by the Wave 1+2 deliverables and the resolved Open Question O.2 of the roadmap.

[1] K. Akama, "An attempt at pregeometry: gravity with composite metric," Prog. Theor. Phys. **60**, 1900 (1978); DOI: 10.1143/PTP.60.1900.

[2] C. Wetterich, "Spinor gravity and diffeomorphism invariance on the lattice," Lect. Notes Phys. **863**, 67 (2013); arXiv:1201.2871; DOI: 10.1007/978-3-642-33036-0_4.

[3] C. Wetterich, "Pregeometry and spontaneous time-space asymmetry," JHEP **06**, 069 (2022); arXiv:2101.11519; DOI: 10.1007/JHEP06(2022)069.

[4] G. E. Volovik, "Fermionic quartet and vestigial gravity," JETP Lett. **119**, 330 (2024); arXiv:2312.09435; DOI: 10.1134/S002136402460006X.

[5] I. García-Etxebarria and M. Montero, "Dai-Freed anomalies in particle physics," JHEP **08**, 003 (2019); arXiv:1808.00009; DOI: 10.1007/JHEP08(2019)003.

[6] Z. Wan and J. Wang, "Beyond Standard Models and Grand Unifications: anomalies, topological terms, and dynamical constraints via cobordisms," JHEP **07**, 062 (2020); arXiv:1910.14668; DOI: 10.1007/JHEP07(2020)062.

[7] M. Kawasaki and T. T. Yanagida, "Dai-Freed anomaly in the standard model and topological inflation," JHEP **11**, 106 (2023); arXiv:2304.10100; DOI: 10.1007/JHEP11(2023)106.

[8] R. N. Mohapatra and A. Y. Smirnov, "Neutrino mass and new physics," Annu. Rev. Nucl. Part. Sci. **56**, 569 (2006); hep-ph/0603118; DOI: 10.1146/annurev.nucl.56.080805.140534.

[9] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro and T. Schwetz, "NuFit-6.0: updated global analysis of three-flavor neutrino oscillations," JHEP **12**, 216 (2024); arXiv:2410.05380; DOI: 10.1007/JHEP12(2024)216.

[10] KamLAND-Zen Collaboration (S. Abe *et al.*), "Search for Majorana neutrinos with the complete KamLAND-Zen dataset," Phys. Rev. Lett. **135**, 262501 (2025); arXiv:2406.11438; DOI: 10.1103/jkf6-48j8.

[11] LEGEND Collaboration (N. Abgrall *et al.*), "LEGEND-1000 Preconceptual Design Report," arXiv:2107.11462 (2021).

[12] J. Tooby-Smith, "HepLean: digitalising high energy physics," Comput. Phys. Commun. **308**, 109457 (2025); arXiv:2405.08863; DOI: 10.1016/j.cpc.2024.109457.

[13] S. Antusch, J. Kersten, M. Lindner and M. Ratz, "Dynamical electroweak symmetry breaking by a neutrino condensate," Nucl. Phys. B **658**, 203 (2003); hep-ph/0211385; DOI: 10.1016/S0550-3213(03)00188-3.

[14] T. Appelquist and J. Carazzone, "Infrared singularities and massive fields," Phys. Rev. D **11**, 2856 (1975); DOI: 10.1103/PhysRevD.11.2856.

[15] R. D. Ball and R. S. Thorne, "The decoupling theorem in effective scalar field theory," arXiv:hep-th/9404156 (1994).

[16] G. F. Giudice, C. Grojean, A. Pomarol and R. Rattazzi, "The strongly-interacting light Higgs," JHEP **06**, 045 (2007); hep-ph/0703164; DOI: 10.1088/1126-6708/2007/06/045.

[17] C. T. Hill, "Bilocal field theory for composite scalar bosons," Entropy **26**, 146 (2024); arXiv:2310.14750; DOI: 10.3390/e26020146.

[18] V. Cirigliano, W. Dekens, J. de Vries, M. L. Graesser and E. Mereghetti, "A neutrinoless double beta decay master formula from effective field theory," JHEP **12**, 097 (2018); arXiv:1806.02780; DOI: 10.1007/JHEP12(2018)097.

[19] S. Navas *et al.* (Particle Data Group), "Review of Particle Physics," Phys. Rev. D **110**, 030001 (2024); DOI: 10.1103/PhysRevD.110.030001.

[20] L. de Moura and S. Ullrich, "The Lean 4 theorem prover and programming language," Lect. Notes Comput. Sci. **12699**, 625 (2021); DOI: 10.1007/978-3-030-79876-5_37.

[21] The Mathlib Community, "The Lean Mathematical Library," Proc. CPP 2020, 367 (2020); DOI: 10.1145/3372885.3373824.